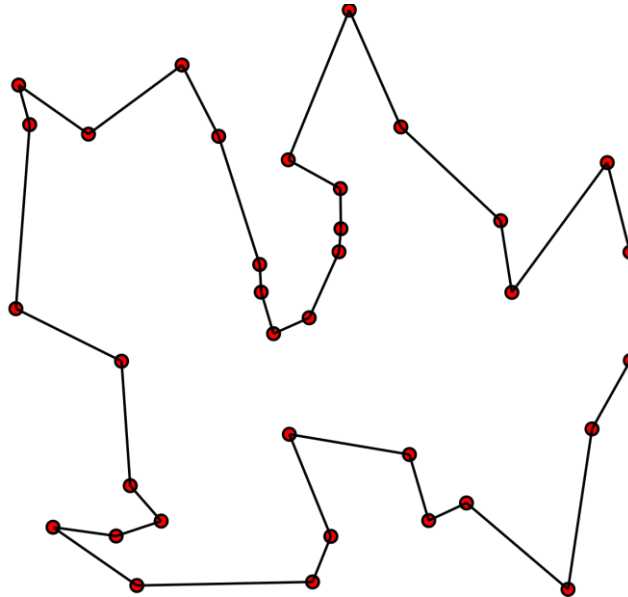


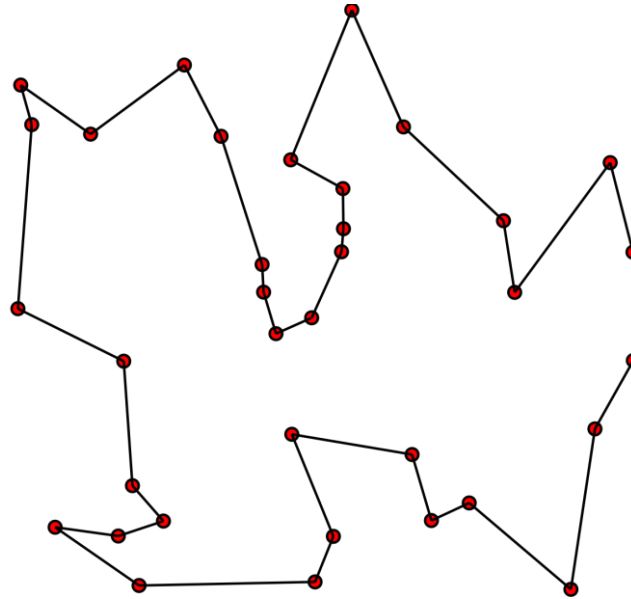
Travelling Salesman Problem



The Problem

Given n cities and the cost c_{ij} of travelling between each pair of cities (i, j) .

Find the cheapest way of visiting all cities and returning to your starting point.



Complexity

Starting point given. How many routes are there?

Let's say we have 4 cities, look at the following two routes

$$A - B - C - D - A$$

$$A - D - C - B - A$$

Do these cost the same or are they different?

They cost the same, therefore we treat them as the same route.

Complexity

From our starting point, how many choices do we have for the next city?

$$3 = 4 - 1$$

For our next city?

$$2 = 3 - 1$$

Thus, we have 3! Possible choices? Are we done?

No, you can go in either direction

$$\begin{array}{l} A - B - C - D - A \\ A - D - C - B - A \end{array}$$

Divide by 2

$$\frac{3!}{2}$$

Complexity

In general, we have to compare $\frac{(n-1)!}{2}$ routes

So, a brute force algorithm will be of order

$$O(n!)$$

That is the time taken for completion increases by approximately $n!$ as n increases!

Ouch!

Some Numbers...

“It would take a laptop 1,000 years to compute the most efficient route between 22 cities.”

Washington Post - Feb 15th , 2018

25,545,471,085,854,720,000 undirected tours for $n = 22$.

This is partly right, what does it assume?

Brute force is the only way to compute this!

NP-Complete

TSP is a member of the set of Non-polynomial Time problems.

The NP bit, means you cannot find the optimal solution in polynomial time, e.g. $O(n^p)$, e.g. $O(n^3)$

The Complete bit means that it is easy to check the solution once you have it.

Sudoku is also NP-Complete.

$$P \neq NP$$

- \$1,000,000 Millennium Problem
- An algorithm that solves TSP in polynomial time would win you \$1,000,000.

Applications

Routing/Timetabling/Scheduling

- Bus Routes
- Service Calls by BT/Sky

Manufacturing Process

- Drilling the holes in a circuit board, in which order does the machine do this?

Florida Sun-Sentinel presented the following article on 20 December 1998

Old Saint Nick has one night to deliver gifts to children around the world. To save time and wear and tear on his sleigh, Santa plots the shortest journey possible among thousands of cities. It sounds easy enough, and for Santa it probably is, but for scientists, who call this the traveling salesman problem (TSP), it is a challenge that grows increasingly difficult as more and more cities are added to the list.

3 If a salesman has to visit three cities, starting from a start, then the shortest possible route is the one with the least number of edges. To figure out the possible routes, let's remove the edges from the graph. Multiple edges between two nodes are treated as a single edge. The route that has the shortest number of edges would be the correct TSP solution. Here are the solutions for three city problems:

[illegible]

4 Another city is added to the set, the possible lines are 24.

The scientists involved like to commission researchers to figure out Santa's route, but even with the computers and the formulas they have worked on for about 10 years, it would take too long to figure out Santa's path. The researchers did manage to compile Santa's daily route through cities around the world.



Computers can solve the problem faster than humans, but they still have to try to connect to a large number of sites. For example, getting how long it would take to get from Indianapolis, Indiana, to New York City to solve this problem.

10 cities	20 cities	30 cities
362,849 possible parish: World Lute 18,500,000 to 20,000,000	1,236 million possible parish: World Lute 151,191,000 to 200,000,000	More than 3,341 billion possible parish: World Lute 280 billion million to 300 billion million

Years ago

In 1954, ENFANT introduced a complete offering of 100,000 shares at a price of \$25.00 per share.

William Cook

University Professor, Combinatorics and Optimization, University of Waterloo

<https://www.math.uwaterloo.ca/~bico/>

Written many books and given excellent presentation on TSP

Well worth a look

Visualisation

[Traveling Salesman Problem Visualization - YouTube](#)

MILP

Let x_{ij} represent whether the edge (i, j) connecting city i and city j and c_{ij} be the cost of that edge.

Then we could try and model the problem as:

$$\begin{aligned} \min. \quad & \sum_{(i,j) \in E} c_{ij} x_{ij} \\ \text{s. t.} \quad & \\ & \sum_{j=1: i \neq j}^n x_{ij} = 1 \quad \forall i \in \{1, \dots, n\} \\ & \sum_{j=1: i \neq j}^n x_{ji} = 1 \quad \forall i \in \{1, \dots, n\} \\ & x_{ij} \in \{0, 1\} \end{aligned}$$

This formulation is wrong? Why?

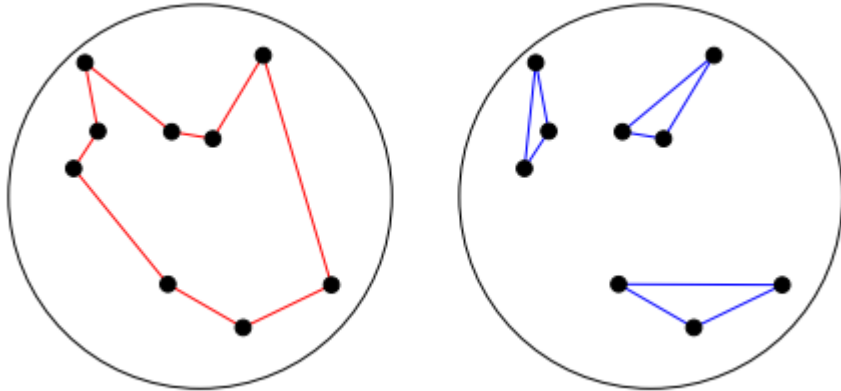
MILP

$$\begin{aligned} \min. \quad & \sum_{(i,j) \in E} c_{ij} x_{ij} \\ \text{s.t.} \quad & \\ & \sum_{j=1: i \neq j}^n x_{ij} = 1 \quad \forall i \in \{1, \dots, n\} \\ & \sum_{j=1: i \neq j}^n x_{ji} = 1 \quad \forall i \in \{1, \dots, n\} \\ & x_{ij} \in \{0, 1\} \end{aligned}$$

Whilst all cities will be visited, it doesn't guarantee a single tour!

We could have lots of subtours that cover all cities.

Subtour example



$$\begin{aligned} \min. \quad & \sum_{(i,j) \in E} c_{ij} x_{ij} \\ \text{s.t.} \quad & \end{aligned}$$

$$\sum_{j=1: i \neq j}^n x_{ij} = 1 \quad \forall i \in \{1, \dots, n\}$$

$$\sum_{j=1: i \neq j}^n x_{ji} = 1 \quad \forall i \in \{1, \dots, n\}$$

$$x_{ij} \in \{0,1\}$$

Both the red (complete tour) and the blue (subtours) have exactly one edge entering and one edge leaving a city!

Solution

Introduce dummy variables $u_i \in R \ \forall i \in \{2, \dots, n\}$ that track the order in which city i is visited.

Then if $x_{ij} = 1$ (that is edge (i, j) is chosen)

- City i is visited before city j
- Hence, we require $u_j > u_i$ which can be written $u_j \geq u_i + 1$

The following constraints do this.

$$u_i - u_j + 1 \leq (n - 1)(1 - x_{ij}) \quad \forall i, j \in \{2, \dots, n\}: i \neq j$$

Why does this work?

The incoming and outgoing constraints mean that for some j
 $x_{ij} = \mathbf{1}$

$$u_i - u_j + 1 \leq 0$$

Which is the same as:

$$u_j \geq u_i + 1$$

Hence selecting edge (i, j) means j visited after i .

This means that $x_{ji} = \mathbf{0}$.

Why, consider if $x_{ji} = \mathbf{1}$, then $u_i \geq u_j + 1$ and $u_j \geq u_i + 1$

This is a contradiction.

Why does this work?

We can continue this reasoning.

We know that there must exist an r_2 such that $x_{1r_2} = 1$ (i.e. we leave city 1).

There also must be $x_{r_2r_3} = 1$, and $x_{r_3r_4} = 1$, and $x_{r_4r_5} = 1$, and ...

Thus, $u_{r_2} < u_{r_3} < u_{r_4} < u_{r_5} < \dots < u_{r_n}$.

Hence there will be a increasing sequence of for u_i , thus precluding $u_i = u_j$.

Why does this work?

Once this sequence is in place.

$$u_{r_2} < u_{r_3} < u_{r_4} < u_{r_5} < \dots < u_{r_n}$$

Then we have to choose some k such that:

$x_{k1} = 1$ i.e. some edge enters city 1.

However, the only remaining city which doesn't have an edge leaving is r_n .

Thus, $k = 1$, i.e., $x_{r_n 1} = 1$

This is valid as we did not include city i in the MTZ constraint. Hence, we have a tour $1, r_2, r_3, r_4, \dots, r_n, 1$

Miller-Tucker-Zemlin (MTZ) Formulation

$$\begin{aligned} \min. \quad & \sum_{(i,j) \in E} c_{ij} x_{ij} \\ \text{s.t.} \quad & \end{aligned}$$

$$\sum_{j=1: i \neq j}^n x_{ij} = 1 \quad \forall i \in \{1, \dots, n\}$$

$$\sum_{j=1: i \neq j}^n x_{ji} = 1 \quad \forall i \in \{1, \dots, n\}$$

$$u_i - u_j + 1 \leq (n - 1)(1 - x_{ij}) \quad \forall i, j \in \{2, \dots, n\}: i \neq j$$

$$\begin{aligned} x_{ij} &\in \{0, 1\} \\ u_i &\in R \end{aligned}$$

Pyomo Code

I have created a demonstration that you can upload to Google Colab.

You will find this on Blackboard.

Tutorial

1. Please use the ipynb notebook to investigate the travelling salesman problem (TSP) -

https://colab.research.google.com/drive/1aGaOpyRexxRXKdu_it68ua8crXVCvNSI?usp=sharing

- You should try out the problem.
- Increase the problem size, when does it become unreasonable to solve?
- Try messing around with the parameters of the genetic algorithm, such as mutation, crossover and population. Can you solve larger TSP problems than the MILP can?

2. I would like you to do some research on the travelling salesman problem.

- Why is it hard to solve?
- What is NP-Hard?
- What is Combinatorial Optimisation

A must watch video

[Math Encounters: "Optimal Tours: The Traveling Salesman Problem" with Bill Cook on August 4, 2021](#)